

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. (CAD/CAM)

Sem - I University Examination, 2010-11

ME – 702 Advances in Manufacturing Technology

Date: 10.05.2011, Tuesday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. Figures to the right indicate **full** marks.
2. Make suitable assumptions and draw neat figures wherever if required.

SECTION-I

- Q1 (a)** Is it possible to weld Magnesium Alloy? If yes, what is the type of welding process used. **03**
- (b)** What are the different shielding gases used in arc welding. **03**
- (c)** Explain the phenomenon of metal transfer in MIG welding with neat sketches. **05**
- Q2 (a)** Discuss the difference between Active and Passive RFID Tags **06**
- (b)** Differentiate between RFID and Barcodes. What is the specific advantage of using RFID system? **06**
- OR**
- Q2 (a)** Explain the phenomenon of Sputtering in PVD. **06**
- (b)** Discuss the type of precursors and its importance in the CVD process **06**
- Q3 (a)** What makes the problem suitable for simulation modeling and analysis? **04**
- (b)** What is the role of modeling and simulation in industrial product design and development? **04**
- (c)** What are the issues in simulating manufacturing systems **04**
- OR**
- Q3 Write Short Notes (Any Three)** **12**
- (a)** Weld Defects and Distortion
- (b)** Friction Stir Welding
- (c)** Welding Power Sources
- (d)** Micro-EDM

SECTION - II

- Q4** Write short notes on (any three) 12
- 1) Characteristics of Hypothesis
 - 2) Methods of Dielectric flushing
 - 3) Hydroforming & its methods
 - 4) Response surface & response surface methodology

- Q5 (a)** Differentiate between Electron Beam and LASER Beam machining by following 06
- (i) principle
 - (ii) setup
 - (iii) applications
- (b)** With schematic of Plasma Arc Machining, explain advantages, limitations and applications. 05

OR

- Q5 (a)** Explain the principle of material removal EDM. What are essential characteristics for Dielectric. 06
- (b)** With neat sketch explain the principle, working and process parameters of Ultrasonic welding. 05
- Q6 (a)** Draw flow diagram for Hypothesis testing. 03
- (b)** What is Regression Analysis? Explain its types. 04
- (c)** The values of x and their corresponding values of y are shown in the table below 05

x	0	1	2	3	4
y	2	3	5	4	6

- a) Find the least square regression line $y = ax + b$.
- b) Estimate the value of y when $x = 10$.

OR

- Q6 (a)** Explain the framework of Virtual manufacturing system showing the structure of virtual manufacturing environment. 06
- (b)** The following table gives the monthly sales (in thousands ₹) of Abhishek Enterprises in three states by its four salesman: 06

States	Salesman			
	A	B	C	D
Haryana	5	4	4	7
Karnataka	7	8	5	4
Rajasthan	9	6	6	7

Set up an ANOVA table for the above information. Calculate F-coefficients and state whether the difference between sales affected by the four salesmen.
Given $F(3,8) = 4.07$.